



DEPI IBM Data Science

Employee Attrition Prediction & Analysis Project

Course instructor:

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Date:

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Project Planning & Management

1. Project Proposal

1.1 Project Overview:

The Employee Attrition Prediction and Analysis project focuses on building a machine learning model to predict employee turnover (attrition) within an organization. By identifying employees who are likely to leave, companies can take proactive measures to improve retention. The project follows a data science lifecycle, from data collection and exploration to model deployment and monitoring, aimed at improving organizational retention strategies.

1.2 Objectives:

1. **Data Preparation:** Collect, explore, and preprocess employee datasets (from open repositories like Kaggle or UCI) to establish a clean foundation for analysis.
2. **Advanced Analysis & Feature Engineering:** Perform in-depth statistical analysis and create advanced features (e.g., tenure categories, salary bands) to uncover key drivers of attrition.
3. **Model Development:** Build, train, and optimize robust classification models (such as Random Forest or XGBoost), ensuring proper handling of class imbalance using techniques like SMOTE.
4. **Deployment & MLOps:** Implement MLOps practices (e.g., MLflow) for tracking, and deploy the final model as an API (Flask/Fast API) coupled with an interactive HR dashboard (Streamlit/Dash) for real-time predictions.
5. **Business Value Presentation:** Deliver comprehensive documentation and an engaging presentation for stakeholders, translating model predictions into actionable HR retention strategies.

1.3 Project Scope:

1. Utilizing open-source employee datasets to train the machine learning models
2. Developing binary classification models to predict attrition vs. non-attrition.
3. Designing an interactive dashboard (e.g., Streamlit or Dash) for HR teams to input data and receive real-time attrition risk scores
4. Deploying the model via a REST API and implementing basic MLOps monitoring.

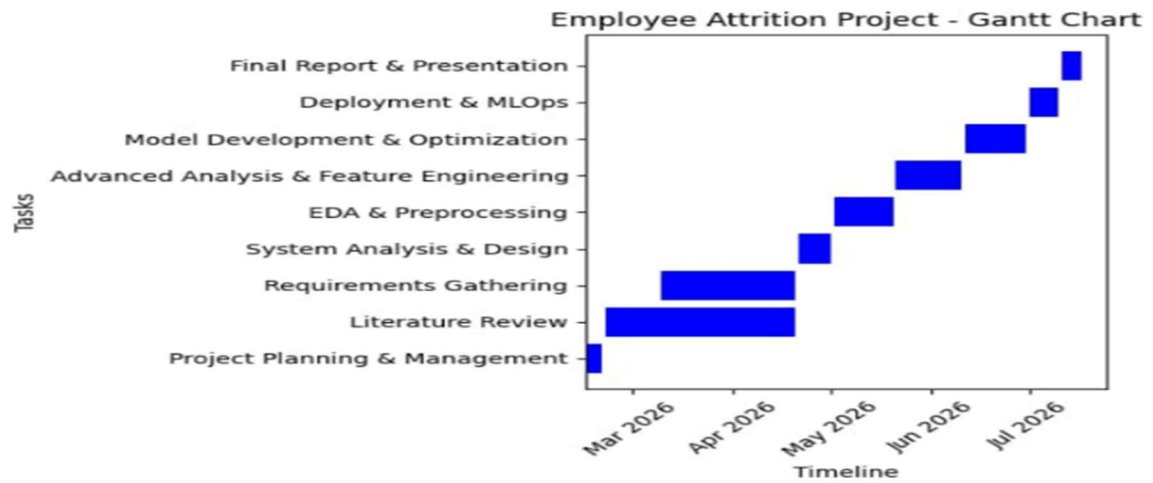
2. Project Planning

This section outlines the project timeline, milestones, deliverables, and resource allocation plan for the Employee Attrition Prediction project.

1.1 Project Timeline & Milestones

Phase	Start Date	End Date	Key Deliverable
Project Planning & Management	15/2/2026	20/2/2026	Proposal & Project Plan
Literature Review	21/2/2026	20/4/2026	Literature Review Report
Requirements Gathering	10/3/2026	20/4/2026	Requirements Document
System Analysis & Design	21/4/2026	1/5/2026	System Design Document
Implementation (EDA + Modeling)	2/5/2026	10/7/2026	Trained & Tuned Model
Deployment & MLOps	1/7/2026	10/7/2026	Deployed API & Monitoring
Final Presentation & Reports	11/7/2026	17/7/2026	Final Report & Presentation

Gantt Chart



1.2 Resource Allocation

Role	Responsibility
Project Manager	Timeline management, coordination, risk tracking
Data Engineer	Data cleaning, preprocessing, data validation
Data Analyst	EDA, visualization, reporting insights
ML Engineer	Model training, tuning, evaluation
MLOps Engineer	Deployment, monitoring, version control
Documentation Lead	Reports, presentation, GitHub documentation

3.Risk Assessment & Mitigation Plan

This section identifies potential risks that may impact the project's execution, evaluate their probability and impact, and define mitigation strategies. Risks will be reviewed bi-weekly during sprint meetings.

Top 5 Risks:

Risk ID	Risk Description	Impact	Probability	Mitigation Plan	Owner
R1	Imbalanced dataset	High	High	Use SMOTE and monitor recall	ML Engineer
R2	Model overfitting	High	Medium	Apply cross-validation and regularization	ML Engineer

R3	Poor data quality	Medium	High	Data cleaning and validation checks	Data Engineer
R4	Deployment failure	High	Medium	Test locally before production deployment	MLOps Engineer
R5	Timeline delay	High	Medium	Weekly monitoring and buffer time	Project Manager

Other Risks:

1- Data & Technical Risks

Even the best algorithm is only as good as the data you feed it.

The "Garbage In, Garbage Out" Factor: If your historical data is messy or missing key details (like why someone left), your predictions will be guessing at best.

Data Leakage: This is a common "oops" in data science. If you include data points that only happen after someone decides to leave (like "unused vacation days" or "exit interview score"), the model will look perfect in testing but fail in real life.

The "Vibe" Problem: Quantitative data (salary, tenure) is easy to track, but qualitative data (a bad boss, burnout, or personal issues) is hard to capture. Your model might miss the human element.

2. Ethical & Privacy Risks

Since you are dealing with people's livelihoods, the stakes are high.

Selection Bias: If the company historically promoted one group over another, the AI might learn that the "neglected" group is more likely to leave. Following the model blindly could reinforce unfair patterns.

Privacy Concerns: Employees might feel "watched." If they know an AI is judging their likelihood of quitting based on their behavior, it can damage trust and actually increase attrition.

The "Self-Fulfilling Prophecy": If a manager sees a high attrition risk score for an employee, they might stop investing in that person's growth, which eventually causes them to quit.

3. Organizational & Action Risks

A model is useless if the company doesn't know how to use the results.

Misinterpretation: A "70% risk" doesn't mean the person is definitely quitting tomorrow. Managers might panic and take the wrong actions.

Lack of Actionable Insight: Knowing who is leaving is great, but if the model doesn't tell you why, you can't fix the problem. Offering a raise to someone who is leaving because of a toxic culture won't work.

Model Decay: People's reasons for quitting change (e.g., the shift to remote work in 2020). If you don't update the model, it will become obsolete quickly.

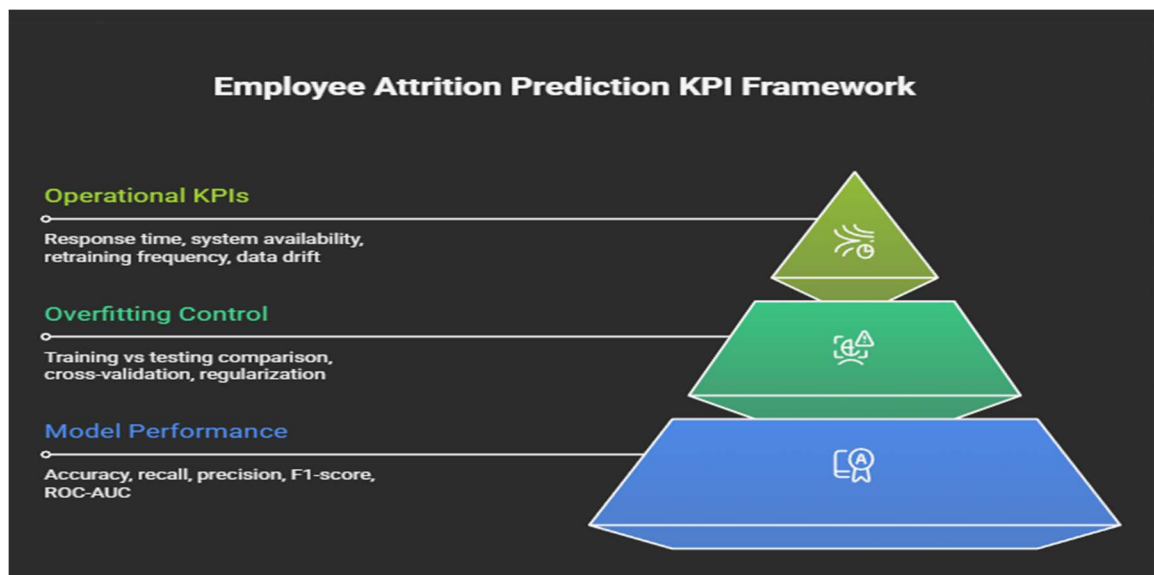
4. Key Performance Indicators (KPIs)

This section defines the Key Performance Indicators (KPIs) used to evaluate the performance and effectiveness of the Employee Attrition Prediction model.

The KPIs focus on measuring model accuracy, generalization capability, and overall predictive strength.

A KPI framework diagram is provided to illustrate the relationship between model performance, operational reliability, and overall system effectiveness.

4.1 KPI Framework Overview



The KPI framework categorizes evaluation metrics into:

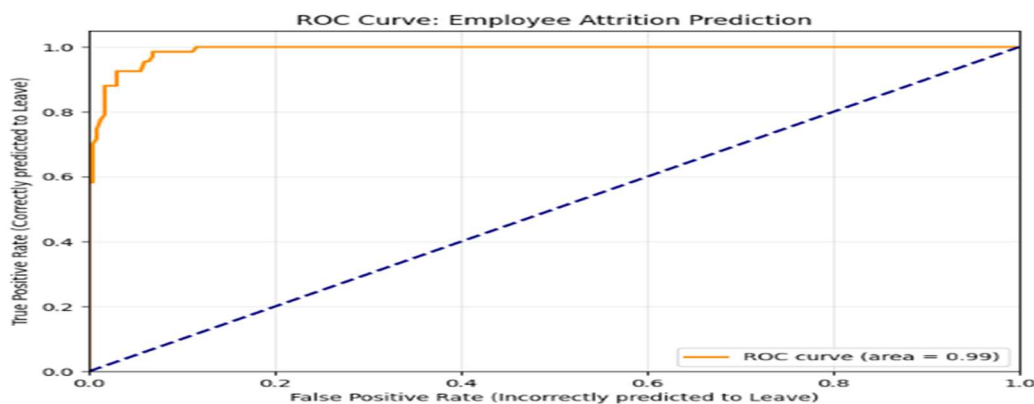
- Model Performance KPIs
- Business-Oriented KPIs
- Operational KPIs

This structure ensures that the system is evaluated from both a technical and practical perspective.

4.2 Model Performance KPIs

The performance of the machine learning models was evaluated using several key performance indicators (KPIs) to ensure accurate and reliable employee attrition prediction.

- **Accuracy:**
Measures the overall percentage of correct predictions made by the model for both attrition and non-attrition classes.
- **Recall:**
Evaluates the model's ability to correctly identify employees who are likely to leave the organization. This metric is especially important in attrition prediction because it helps minimize false negatives.
- **Precision:**
Measures the proportion of correctly predicted attrition cases out of all cases predicted as attrition by the model.
- **F1-Score:**
Provides a balanced evaluation by combining both Precision and Recall into a single metric, making it useful when dealing with imbalanced datasets.
- **ROC-AUC Score:**
Measures the model's ability to distinguish between employees who will leave and those who will stay. Higher AUC values indicate stronger classification performance and better model reliability.



The ROC curve evaluates the model's ability to distinguish between employees who are likely to leave (Attrition = 1) and those who will stay (Attrition = 0).

The model achieved an AUC score of **0.99**, which indicates excellent classification performance and very strong discriminative capability.

This means the model can effectively separate high-risk employees from low-risk employees across different classification thresholds.

However, further validation is necessary to ensure that this high performance is not due to overfitting.

4.3 Overfitting Control

To ensure proper generalization:

- The difference between training and testing performance is monitored.
- Cross-validation techniques are applied.
- Regularization methods are used where necessary.

This ensures that the model performs consistently on unseen data.

4.4 Operational KPIs

Although the project focuses on machine learning performance, operational reliability is also considered:

- Prediction response time
- System availability
- Model retraining frequency
- Data drift monitoring

These ensure long-term stability of the prediction system.

2. Literature Review

This section provides a comprehensive review of existing research and methodologies related to employee attrition prediction, machine learning classification techniques, and HR analytics. The literature review informed the design decisions, model choices, and feature engineering strategies adopted in this project.

2.1 Employee Attrition: Definition and Business Impact

Employee attrition refers to employees leaving an organization through resignation, retirement, or other forms of departure. High attrition negatively affects productivity, organizational stability, and recruitment costs. Research shows that employee turnover is strongly influenced by factors such as job satisfaction, career growth opportunities, management relationships, compensation, and work-life balance. Because replacing employees is expensive, predicting attrition has become an important business objective.

2.2 Machine Learning Approaches in Attrition Prediction

Machine learning models are widely used to predict employee attrition. Studies have shown that ensemble models such as Random Forest and XGBoost achieve strong predictive performance because they can capture complex relationships between employee features. Logistic Regression is also commonly used because it provides interpretable results that help HR teams understand why employees may leave the organization.

2.3 Handling Class Imbalance in Attrition Datasets

Attrition datasets are usually imbalanced because only a small percentage of employees leave the company. To address this issue, techniques such as SMOTE (Synthetic Minority Oversampling Technique) are applied to generate additional minority-class samples and improve model performance. Research indicates that combining resampling methods with ensemble learning models produces better prediction results on imbalanced HR datasets.

2.4 Feature Engineering for HR Analytics

Feature engineering plays a major role in improving attrition prediction accuracy. Previous studies demonstrated that creating derived features such as tenure categories and salary bands enhances model performance compared to using raw data only. Feature selection techniques like Recursive Feature Elimination (RFE) are also effective for reducing unnecessary variables while preserving predictive power.

2.5 Dashboard and Deployment for HR Decision Support

For HR teams to benefit from predictive models, results must be accessible and easy to interpret. Interactive dashboards built using tools such as Streamlit or Dash allow HR managers to monitor employee attrition risk in real time. Deployment frameworks like Flask or FastAPI enable integration with HR systems, while MLflow supports experiment tracking, model versioning, and reproducibility.

2.6 Overall Conclusion

The literature supports using ensemble machine learning models, SMOTE-based imbalance handling, and feature engineering techniques for employee attrition prediction. In addition, interpretability methods and interactive dashboards are essential for helping HR departments make informed decisions and take proactive actions to reduce employee turnover.

Requirements Gathering and System Analysis & Design

1. Problem Statement & Objectives

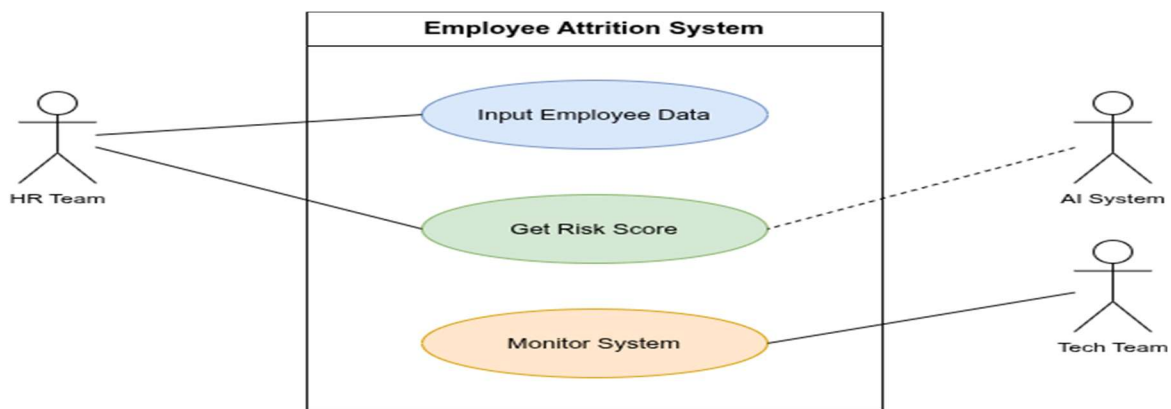
1.1. Problem Statement:

The Employee Attrition Prediction and Analysis project focuses on building a machine learning model to predict employee turnover (attrition) within an organization. By identifying employees who are likely to leave, companies can take proactive measures to improve retention

1.2. Objectives:

- Collect, explore, and preprocess employee datasets (from open repositories like Kaggle or UCI) to establish a clean foundation for analysis.
- Perform in-depth statistical analysis and create advanced features (e.g., tenure categories, salary bands) to uncover key drivers of attrition.
- Build, train, and optimize robust classification models (such as Random Forest or XGBoost), ensuring proper handling of class imbalance using techniques like SMOTE.
- Implement MLOps practices (e.g., MLflow) for tracking and deploy the final model as an API (Flask/Fast API) coupled with an interactive HR dashboard (Streamlit/Dash) for real-time predictions.

2. Use Case Diagram & Descriptions



3. Functional & Non-Functional Requirements

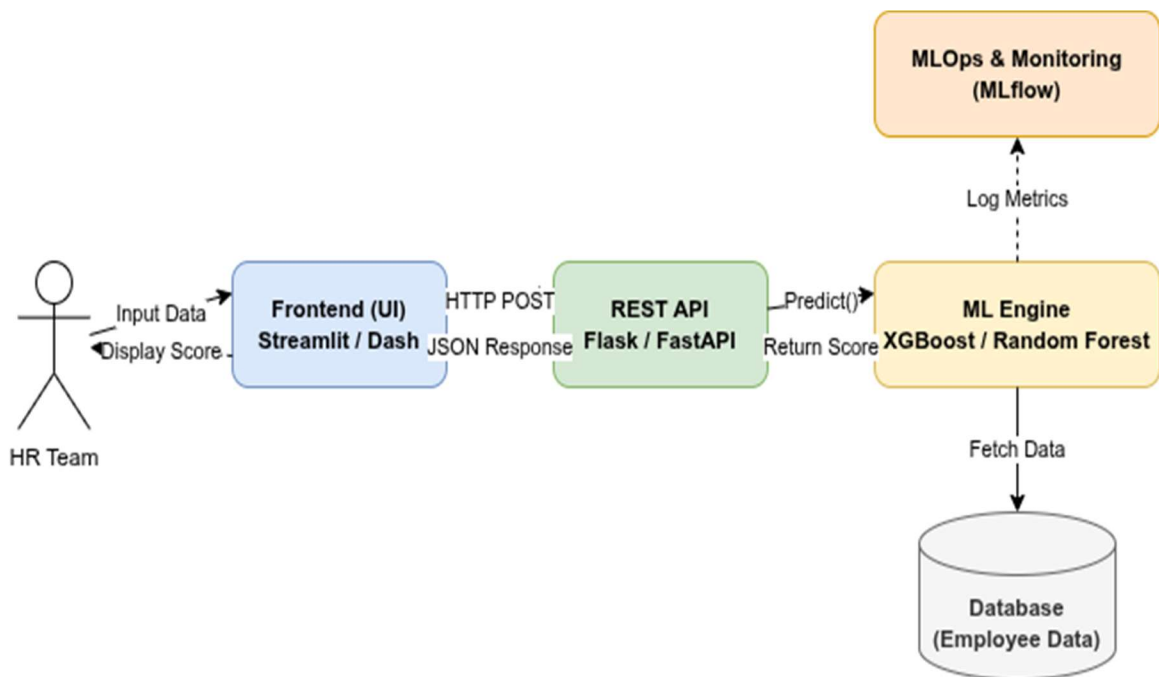
3.1. Functional Requirements:

- The system must feature an interactive dashboard (e.g., Streamlit or Dash) for HR teams to input data.
- The system must utilize binary classification models to predict attrition vs. non-attrition.
- The system must be capable of delivering real-time attrition risk scores.
- The predictive model must be deployed via a REST API.

3.2. Non-Functional Requirements:

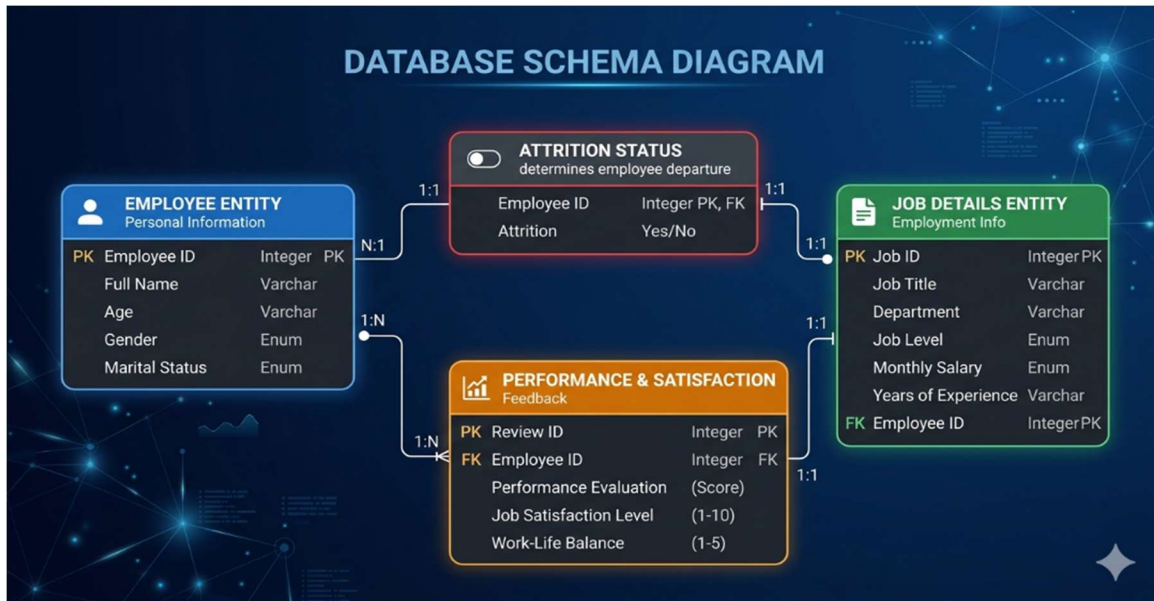
- **Performance:** The system must be monitored for efficient response time and consistent system availability.
- **Reliability:** The model must maintain control overfit through training vs testing comparison, cross-validation, and regularization.
- **Accuracy:** The model's effectiveness must be measurable using predefined KPIs including Accuracy, recall, precision, F1-score, and ROC-AUC.
- **Maintainability:** The system requires basic MLOps monitoring to ensure long-term stability and tracking.

4. Software Architecture



2. Database Design & Data Modeling

ER Diagram (Entity-Relationship Diagram)



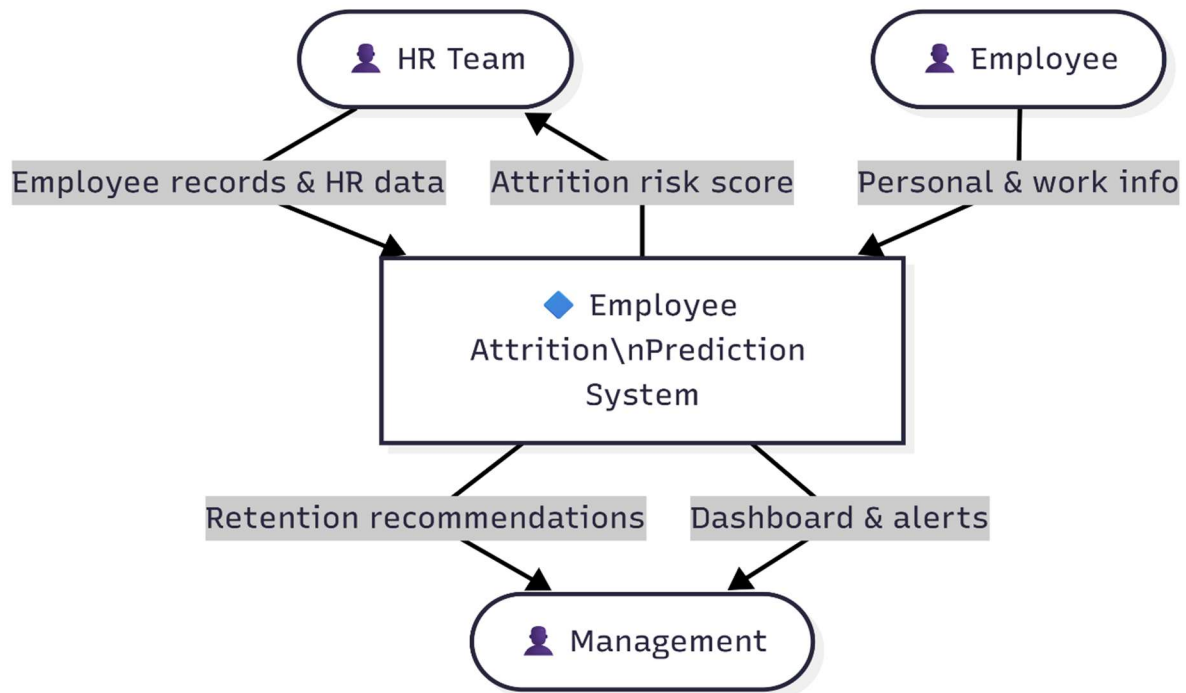
This design uses a normalized relational structure to organize data efficiently. And these are the factors structuring this design:

- **Zero Redundancy:** By ensuring features like "Salary" or "Age" appear in only one table, you eliminate data conflicts and save storage space.
- **Predictive Readiness:** Isolating the Attrition Status creates a clear "Target Variable" (Yes/No), making it much easier to feed this data into a Machine Learning model.
- **Logical Grouping:** Employee Entity: Handles identity (Who they are).
- **Job Details:** Handles professional context (What they do).
- **Performance:** Handles feedback (How they feel/act).
- **Scalability:** The "One-to-Many" link for Performance allows you to store multiple years of evaluations for a single employee without duplicating their personal info.
- **Data Security:** You can restrict access to the "Salary" or "Personal" tables while still letting managers view "Performance" data.

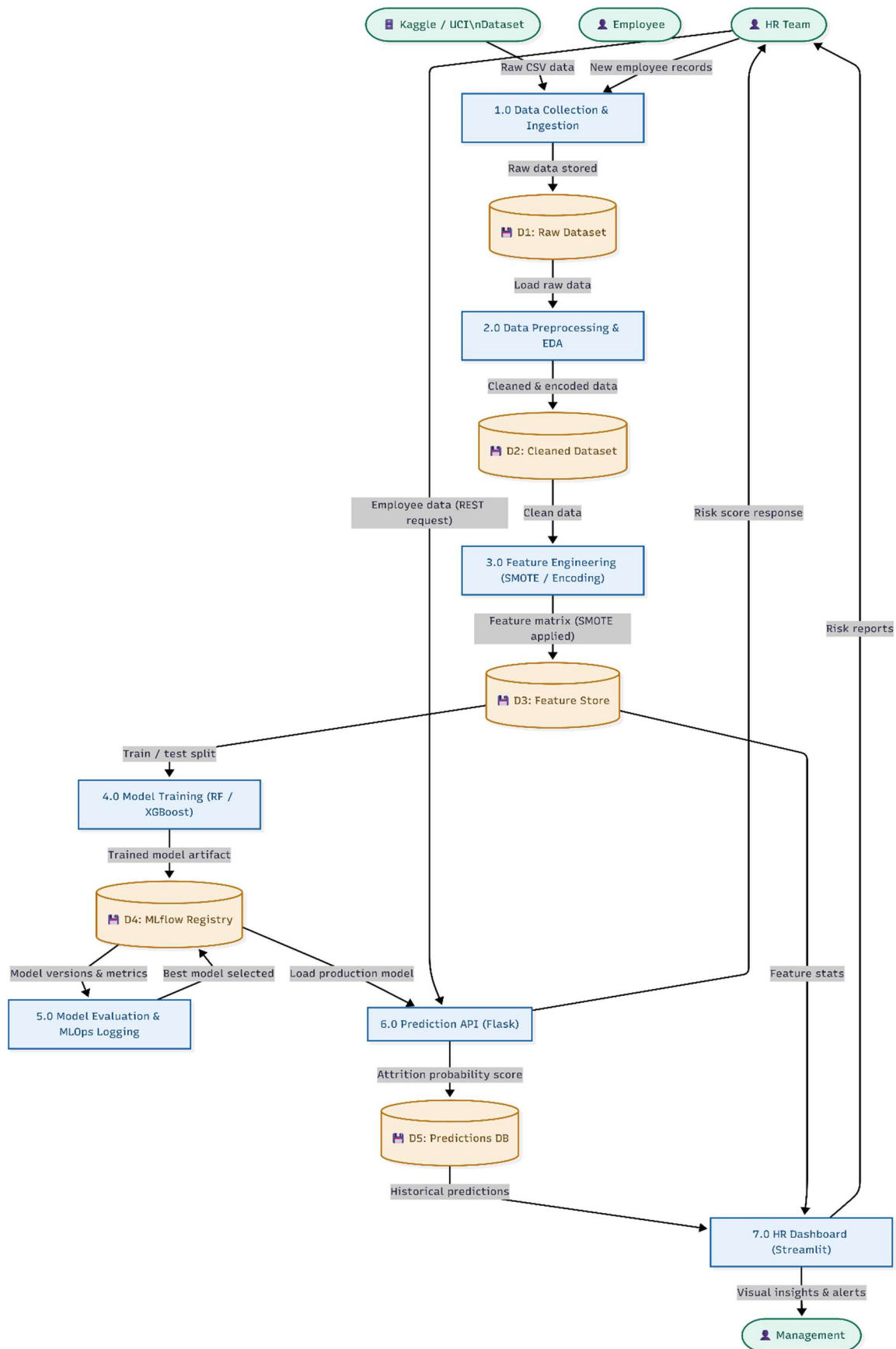
3. Data Flow & System Behavior

DFD (Data Flow Diagram) – Context-level and detailed levels showing how data moves through the system.

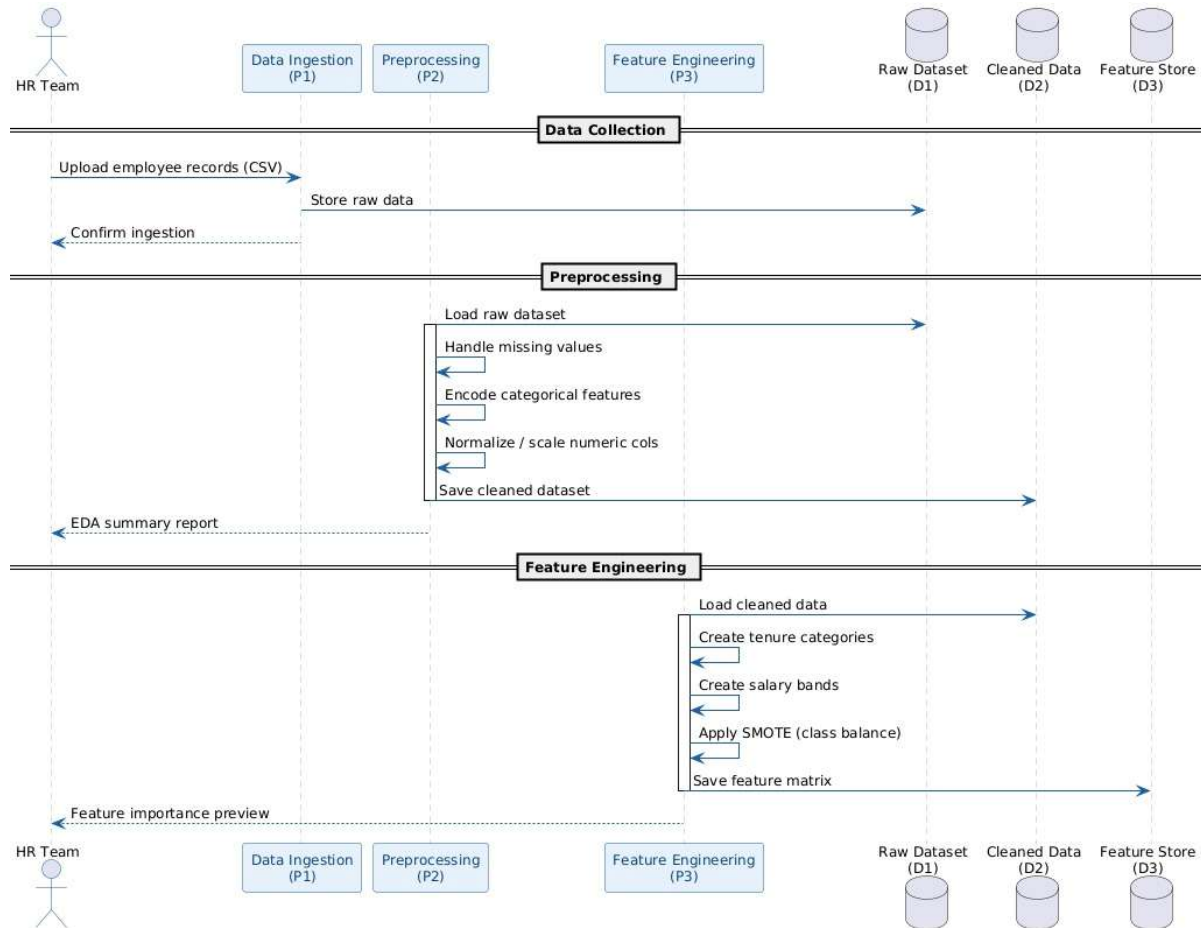
Context-level



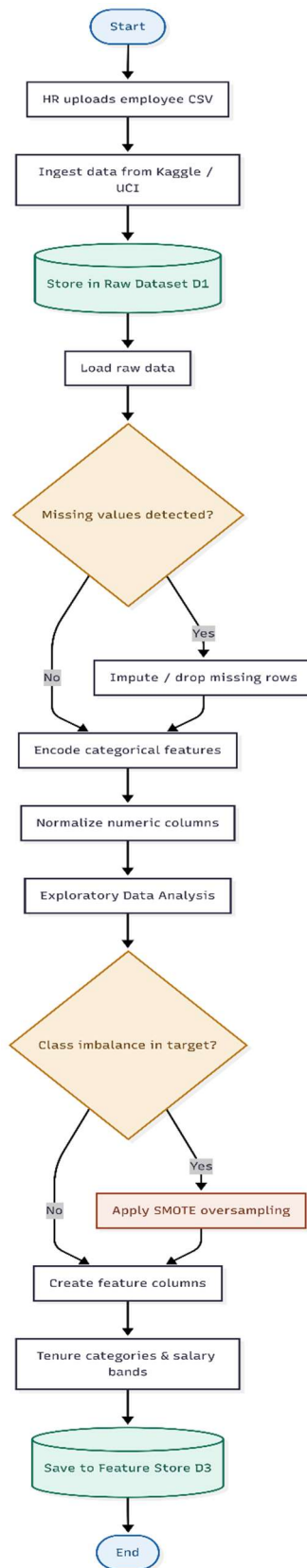
Detailed levels



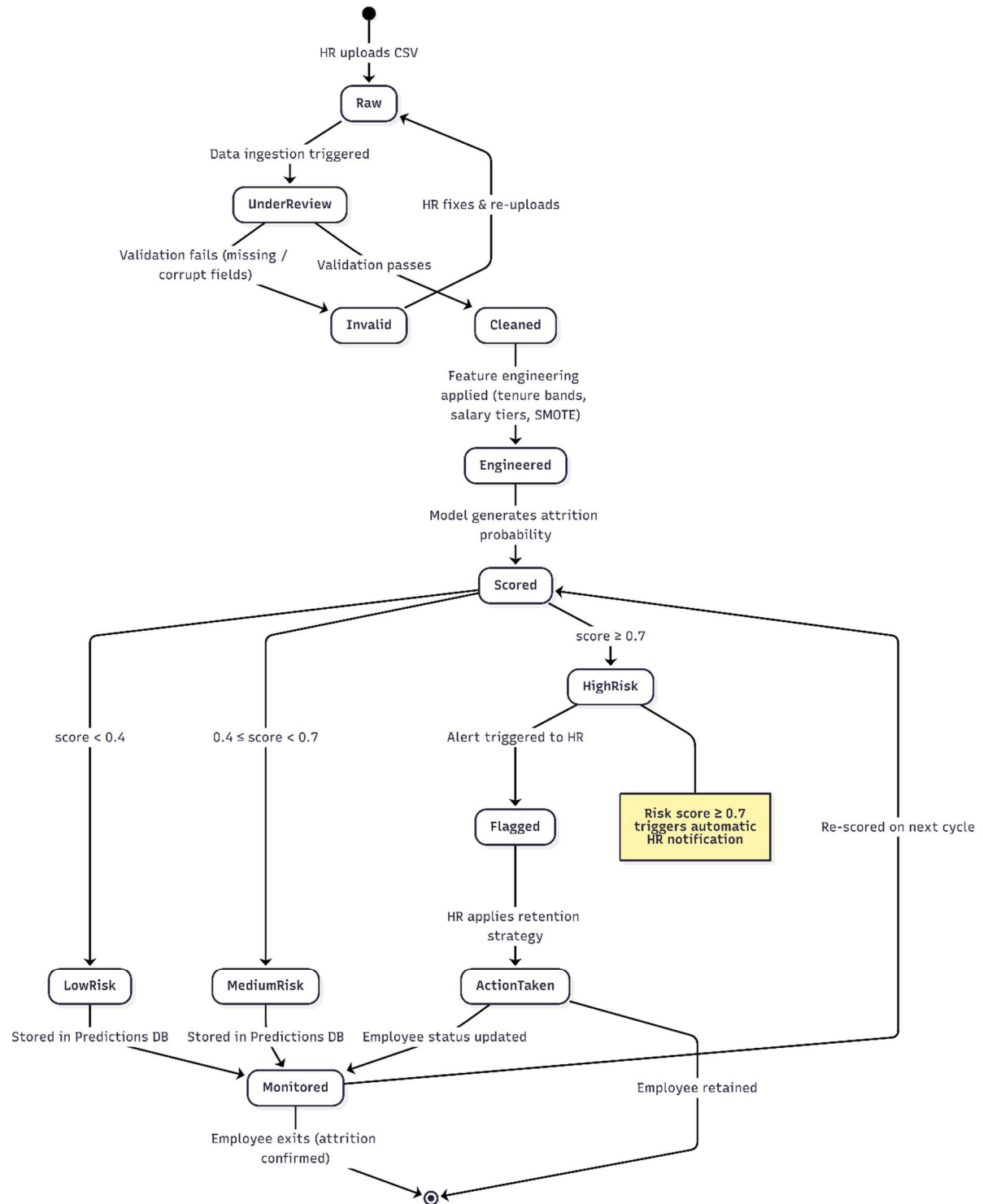
Sequence Diagrams – Process flow representation of key interactions between components.



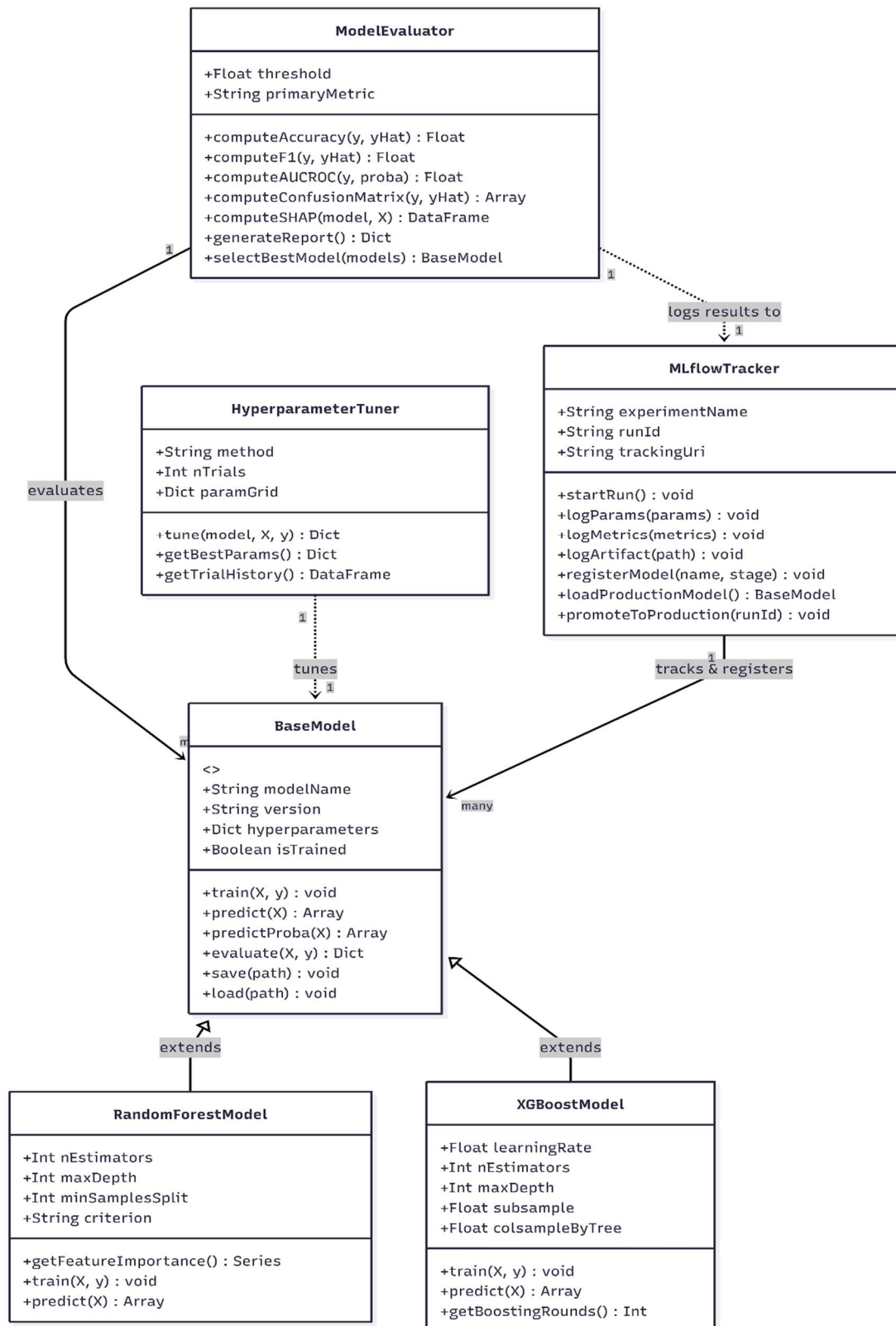
Activity Diagram – Visualizing the workflow of processes or user actions within the system.



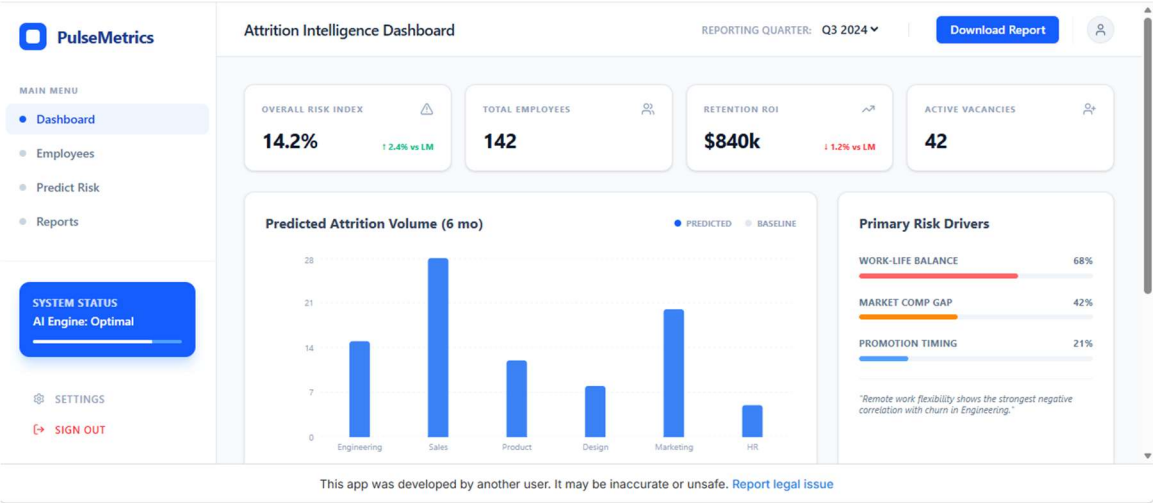
State Diagram – Represents different states of an object and how it transitions between them.

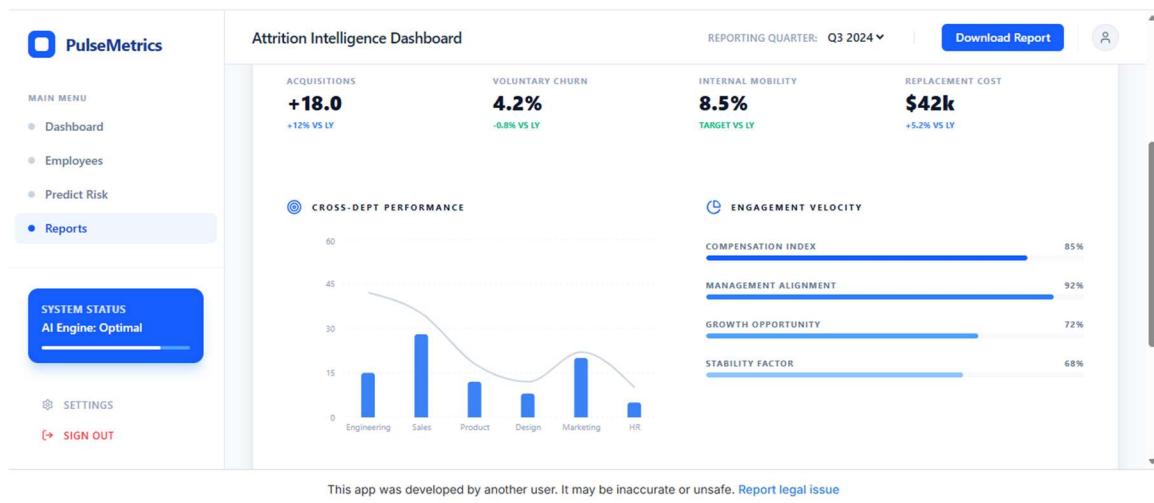


Class Diagram – Defines the structure of the system by showing classes, attributes, methods, and relationships.



4. UI/UX Design & Prototyping





5. Deployment System & Integration

5.1 Technology Stack

The Employee Attrition Prediction system uses modern technologies for machine learning, deployment, and visualization:

Layer	Technologies	Purpose
Machine Learning	Python, Scikit-learn, XGBoost, SMOTE	Model training and prediction
Backend / API	Flask or Fast API	REST API for real-time predictions
Frontend / Dashboard	Streamlit or Dash	Interactive HR dashboard
Database	PostgreSQL or SQLite	Store employee and prediction data
MLOps	MLflow	Model tracking and versioning
Version Control	Git & GitHub	Code management and CI/CD

5.2 Deployment Diagram

The system follows a **three-tier architecture**:

1. Data Layer

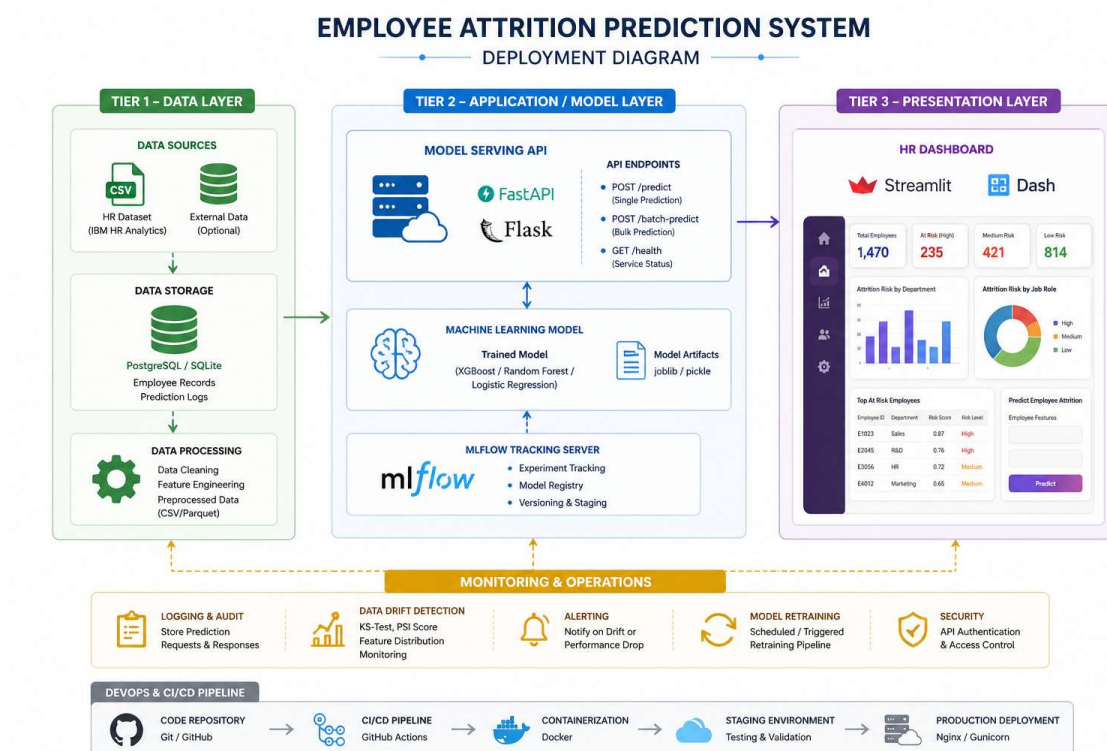
- Stores employee data in PostgreSQL/SQLite.
- Handles preprocessing and feature-engineered datasets.

2. Application / Model Layer

- Trained models are saved using Joblib/Pickle.
- Flask/FastAPI provides prediction APIs such as:
 - /predict
 - /batch-predict
 - /health
- MLflow tracks experiments and model versions.

3. Presentation Layer

- Streamlit/Dash dashboard visualizes attrition risk.
- HR managers can view analytics and predict employee attrition in real time.



5.3 Main System Components

The system contains the following core components:

1. **Data Ingestion**
Loads and validates employee datasets.
2. **Preprocessing & Feature Engineering**
Applies encoding, scaling, and feature generation.
3. **Model Training**
Trains ML models, applies SMOTE, and evaluates performance.
4. **Model Serving API**
Provides prediction services through REST endpoints.
5. **Dashboard**
Displays attrition insights and prediction results for HR teams.
6. **Monitoring System**
Tracks predictions, logs activity, and detects data drift.

5.4 Deployment Strategy

The deployment pipeline includes:

- **Local Development & Testing:**
Build and test the system locally using Docker.
- **Staging Environment:**
Deploy to a testing server and run CI/CD validation.
- **Production Deployment:**
Deploy the API and dashboard for HR team usage.
- **Model Retraining:**
Periodically retrain the model when performance decreases or data drift is detected.